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24 June 1954
SUPERSEDING
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MILITARY SPECIFICATION

CONTROLLER, DRIVE, M12AI and M12E2

1. SCOPE

1.1 This specification covers two drive controllers which are used in conjunction with oil gears to control the positioning of turrets on 401mm Antiaircraft Guns. Drive Controller M12A1 is a component of System, Local Control, M22; Drive Controller M12E2 is a component of System, Local Control, M16A1E1.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

U. S. Army
51-7 Instruments, Fire Control, General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps
F8202433 Controller, Drive, M12Al
F7665397 Controller, Drive, M12E2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting *officer).*

3. REQUIREMENTS

3.1 General. - Unless otherwise specified, the requirements common to all fire control instruments as contained in Specification 51-? shall be mandatory.

3.2 **Material. - Material shall be in accordance with the applicable drawings, and specifications forming a part of this specification.**

3.3 **Electrical**

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3. 3.1 **Voltage input.** - The drive controller shall meet the performance requirements of this specification when 115 volts, 60 cycles, alternating current is applied, The instrument shall **not be damaged when 130 volts,** 60 cycles is applied for a period of two hours.

3.3.2 **High potential. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength of the insulation sufficient to cause failure of the drive controller when it is subjected in the test specified in 4.2.1.**

3. 3.3 Insulation resistance. - All drive controllers shall have a minimum insulation resistance of 5 megohms when subjected to the test specified in 4.2.2,

3.4 Performance

3.4.1 Rate switch. - When the rate motor switch is in the "OFF" (down) position, the rate motor shall be de-energized and when the switch is in the "ON" (up) position, the rate motor shall be energized.

3.4.2 Firing circuit. - When either or both triggers are compressed, and the safety switch is in the "Oh?' position as noted on the switch name plate, continuity shall be demonstrated through the firing circuit. When the firing safety switch is in the "OF F" position, the firing circuit shall not be energized.

3.4.3 **Handgrip displacement. - Handgrips of each drive controller shall operate smoothly about either axis and shall return to neutral when released at any position.** When the handgrips are displaced 3 to 5 degrees from the neutral position; and then released, the transmitter speed shall not exceed one revolution per minute. This limit shall apply to both directions of rotation in azimuth and elevation.

3.4.4 Output speed. - When the drive controller is adjusted and mounted properly, and the handgrips are displaced to their maximum limits in azimuth and elevation, the minimum output speeds of the transmitters shall be as follows:

Drive controller	Azimuth (rpm)	Elevation (rpm)
M12A1	90	69
M12E2	117	71

3.5 Environmental

3.5.1 **Temperature. - The drive controller shall meet the performance requirements of this specification throughout the air temperature range extending from plus 51. 5" C (+ 125° F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr.), to minus 31. 6° C (- 25°F) (minimum exposure of 3 days without benefit of solar radiation). The drive controller shall be capable of safe storage within the air temperature range extending from plus 71° C (+ 1600 F) (for periods as long as 4 hours per day) to minus 62.2° c (- 80° F) (for periods of at least 3 days duration). Test shall be conducted in accordance with 4.3.2.**

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3.5.2 Humidity. The drive controller shall meet the performance requirements of this specification immediately after being subjected to the humidity test specified in 4.3.30

3.5*3 Rain. There shll be no evidence of water penetration or damage to the= controller after it has been subjected to the rain test in accordance with 4.3.4.

3.5.4 Vibration. - The drive controller shall not suffer damage or fail to meet the performance requirements of this specification immediately after being subjected to the vibration test specified in 4.3.5.

3.5.5 Shock. - The drive controller shall not suffer damage or fail to meet the performance requirements of this specification immediately after being subjected to the shock test specified in 4.3.6.

3.6 Workmanship.- Workmanship shall be in accordance with Specification 51-7.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - Unless otherwise specified, each drive controller shall be inspected conformance with this specification and Specification 51-7. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60° F to 90° F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Electrical

4.2.1 High potential. - The instrument shall be subjected to a potential of 900 volts with a frequency of 60 cycles per second for a period of one minute. Three of the first five and one out of every additional 50 or fraction thereof shall be subjected to the test voltage which shall be applied at a rate not exceeding 500 volts (-s) per second. For all other units the test voltage may be applied instantaneously for a period of not less than 5 seconds or more than 30 seconds. Rate motor and capacitor shall be disconnected for these tests to prevent damage to these components. Insulation resistance test shall be performed immediately after this test.

4.2.2 Insulation resistance. - The insulation resistance shall be measured by an approved method using a source of d, c. potential of 500 volts. The rate motor and capacitor shall be disconnected for this test.

4.3 Environmental

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4.3.1 General. -The environmental tests specified shall be limited to those contained in this paragraph. Three of the first five drive controllers and one out of every additional one hundred or fraction thereof shall be tested

4.3.2 Temperature. - The drive controller shall be exposed to an air temperature of minus 62.2° C (-800 F) for a period of 3 days. The air temperature shall then be raised to minus 31. 6°C (-25°F) and after the internal temperature of the unit has reached equilibrium (approximately 4 hours), it shall be tested for conformance with the performance requirements of this specification. The air temperature shall be raised to plus 71° C (+160° F) and after the internal temperature of the unit has reached equilibrium at that temperature it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5°C (+ 125° F) and after the internal temperature of the unit has reached equilibrium at that temperature, the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The unit then shall be tested for conformance with the performance requirements of this specification.

4.3.3 Humidity.- A test chamber that is capable of being sealed shall be used; The source of heat shall be arranged in such a manner that the radiant heat will not fall upon the drive controller being tested. A unit shall be placed in the chamber, the temperature of which is adjusted to plus 46.1° C ± l.l° C (+ 115° F 22°F) and the relative humidity adjusted to 95:5 percent. The unit shall remain under the above conditions for a period of 6 hours. The temperature of the chamber shall be readjusted to plus 26.7°C ±1.1 °C (+ 80° F ±2° F) and the unit shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated ten times. Immediately after this test the drive controller shall meet the requirements of 3.5.2.

4.3.4 Rain - The drive controller shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C to 30°C(68° F to 86° F) throughout the test period. A simulated rainfall of 4 + 1 inch per hour shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between ll° C to 24°C (51.8° F to 75.2° F). The water spray shall be direction the drive controller for a period of 2 hours in directions ranging from vertical to 45° from the vertical. The simulated rain shall be directed on all surfacesof the drive controller that are normally subjected to rain when the instrument is in its operating position.

4.3.5 Vibration. - The drive controller shall be mounted rigidly using its own mounting provisions, to simulate field use. Each unit tested shall be vibrated successively along each of the three mutually perpendicular directions with accelerations and excursions as shown below. Applied frequency shall be changed at a uniform rate between the limits indicated, the rate of change shall not exceed one cycle per second.

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Frequency limits	Excursion	Accreleration	Time (each direction}
0-10 cps	0.060 inch	———	l hour
10 -20 Cps	0. 030 inch	———	2 hours.
20-400 Cps		4.5 g's	2 hours

4.3.6 Shock - The drive controller shall be mounted rigidly, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions, making a total of 18 shocks. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock pulse.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking - Packaging, packing, and marking shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data. - Contracts and invitation for Bids should specify the title, number, date of this specification and type of packing.

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